Dynamic Systems Forecasting
Challenge Providers

qBraid MITRE JonesTrading

GLOBAL INDUSTRY CHALLENGE 2026

Global Industry Challenge 2026

Phase 2 Challenge Description

CHALLENGE PROVIDER:
qBraid, MITRE, and JonesTrading

FOCUS AREA:
Dynamic Systems Forecasting

TITLE:
Quantum Reservoir Computing for Time-Series Intelligence

1. INTRODUCTION

Welcome Message

Welcome to the qBraid Global Industry Challenge for Quantum World Congress 2026. This Challenge invites participants to design and benchmark Quantum Reservoir Computing (QRC) systems for forecasting complex, chaotic time-series data, with industry-relevant use cases in financial volatility prediction and weather forecasting. The Challenge is hosted on the qBraid platform and supported by partner organizations MITRE and JonesTrading.

Overview of the Organization

qBraid is a quantum computing software company headquartered in Chicago, IL. The qBraid platform provides centralized access to 24+ quantum devices, cross-framework conversion across 18+ quantum SDKs, and a managed cloud development environment used by 25,000+ researchers worldwide. qBraid serves as Challenge sponsor and provides the technical platform on which all submissions will be developed and evaluated.

MITRE is non-profit organization that operates federally funded research and development centers (FFRDCs) supporting the U.S. government across defense, intelligence, aviation, healthcare, cybersecurity, and emerging technologies. Headquartered in McLean, VA, MITRE works at the intersection of public and private sectors to solve complex, mission-critical challenges. The organization has a growing focus on quantum computing, artificial intelligence, and advanced analytics, applying these technologies to real-world systems such as national security, infrastructure resilience, and large-scale data modeling. In this Challenge, MITRE contributes expertise in complex systems, modeling, and evaluation frameworks to help guide the development and benchmarking of quantum reservoir computing approaches.

JonesTrading is a leading institutional brokerage and capital markets firm headquartered in New York, NY. The firm provides trading, research, investment banking, and capital markets advisory services to institutional clients, with a strong focus on data-driven strategies and advanced analytics. JonesTrading has been actively exploring the application of emerging technologies—including quantum computing and machine learning—to financial markets, particularly in areas such as volatility modeling, liquidity forecasting, and complex systems analysis. In this Challenge, JonesTrading brings domain expertise in financial time-series data and market dynamics, helping to shape real-world use cases and evaluate the practical impact of quantum-enabled forecasting solutions.

2. THE CHALLENGE

Problem Statement

Forecasting chaotic, nonlinear time-series is one of the foundational hard problems in machine learning and applied mathematics. Two important real-world instances are financial market volatility and atmospheric weather: both are governed by high-dimensional nonlinear dynamics, exhibit regime shifts and extreme events, and resist accurate prediction even with state-of-the-art classical methods.

Quantum Reservoir Computing (QRC) has emerged as one of the most promising near-term quantum machine learning paradigms for time-series problems. Unlike variational quantum algorithms, QRC is gradient-free, avoids barren plateaus, and exploits the natural high-dimensional dynamics of a quantum system as a feature extractor. Recent experimental work has demonstrated QRC at scales beyond what variational methods can reach on current hardware, with competitive performance on classification and forecasting tasks.

This Challenge asks participants to design, justify, and benchmark QRC architectures for two industry-relevant forecasting tracks. Teams will select one track and use publicly available datasets to make a rigorous technical case for their approach.

Background and Context

Reservoir computing reframes time-series learning by fixing a high-dimensional nonlinear dynamical system (the "reservoir") and training only a linear readout on its state trajectory. In the quantum case, the reservoir is a controllable quantum system, for example a transverse-field Ising chain, a Rydberg atom array, or a cavity QED system, whose Hilbert space provides exponentially many degrees of freedom for feature extraction. Recent results have shown:

- QRC can be scaled to over 100 qubits on current analog quantum hardware, far beyond what variational algorithms can reach (Kornjača et al., 2024).
- Few-atom QRC systems with feedback can match or exceed classical reservoir computing on Mackey–Glass and waveform classification tasks (Zhu et al., 2025).
- Carefully designed QRCs can robustly forecast chaotic dynamics and capture invariant properties such as Lyapunov spectra (Ahmed et al., 2025).
- QRC has been applied to realized volatility forecasting and aerospace material degradation, demonstrating relevance to concrete industry problems (Li et al., 2025; Tandon et al., 2025).

Despite this progress, fundamental questions remain open: what reservoir Hamiltonians and encoding strategies are best matched to a given signal class, how does noise interact with reservoir expressivity, and where does QRC genuinely compete with strong classical baselines such as Echo State Networks, LSTMs, and GARCH-family models. This Challenge is structured to push teams to engage these questions directly.

Desired Outcomes
By the end of the Challenge, successful teams will have:

1. Designed a QRC architecture (Hamiltonian, input encoding, readout, and any feedback mechanisms) with explicit theoretical or analytical justification for the choices made.
2. Benchmarked the architecture against strong classical baselines on a publicly available dataset in either the financial or weather forecasting track.
3. Characterized how performance scales with reservoir size, encoding density, shot budget, and noise.
4. Produced a fully reproducible workflow on the qBraid platform that judges can re-execute without modification.

3. RESOURCES AVAILABLE

Key Datasets
Teams must use publicly available datasets for Phase 2. Suggested sources include, but are not limited to:

Track A: Financial Volatility

- Yahoo Finance and Kaggle (free historical equity OHLCV data at daily and intraday resolutions).
- Oxford-Man Institute Realized Volatility Library (realized variance and related estimators for major equity indices).
- FRED (Federal Reserve Economic Data) for macroeconomic time-series.

Track B: Weather and Atmospheric Forecasting

- NOAA Integrated Surface Database (ISD) and ASOS station observations: hourly temperature, pressure, humidity, wind.
- ECMWF ERA5 reanalysis (open access) and ECMWF Open Data forecasts.
- NOAA Global Forecast System (GFS) public archives.

Software and Tools
qBraid Lab. Cloud-based development environment with managed Python/Julia kernels, pre-installed quantum SDKs (Qiskit, Cirq, PennyLane, Bloqade, Braket SDK), and direct access to simulators and quantum hardware.
Documentation: https://docs.qbraid.com

qBraid SDK. Cross-framework runtime that allows teams to write provider-agnostic code and target multiple backends without rewriting their workflow.

QuEra QRC-tutorials. Open-source Bloqade-based notebooks that reproduce the Kornjača et al. work and demonstrate Aquila job submission, available at github.com/QuEraComputing/QRC-tutorials. A useful starting point for teams targeting analog neutral-atom hardware.

A full list of available simulators and quantum hardware backends is provided in the Phase 2 Requirements section below.

Background References

Teams should ground their design choices in the published literature. Recommended starting points:

1. Kornjača, M. et al. “Large-scale quantum reservoir learning with an analog quantum computer.” arXiv:2407.02553 (2024). [108-qubit QRC on neutral-atom hardware.]
2. Zhu, C., Ehlers, P. J., Nurdin, H. I., Soh, D. “Practical few-atom quantum reservoir computing.” Phys. Rev. Research 7, 023290 (2025). arXiv:2405.04799. [Minimalistic cavity-QED QRC with feedback.]
3. Ahmed, O., Tennie, F., Magri, L. “Robust quantum reservoir computers for forecasting chaotic dynamics: generalized synchronization and stability.” Proc. R. Soc. A 481, 20250550 (2025). arXiv:2506.22335.
4. Li, Q., Mukhopadhyay, C., Bayat, A., Habibnia, A. “Quantum Reservoir Computing for Realized Volatility Forecasting.” arXiv:2505.13933 (2025). [Direct application to financial forecasting.]
5. Tandon, A. et al. “Quantum Reservoir Computing for Corrosion Prediction in Aerospace: A Hybrid Approach for Enhanced Material Degradation Forecasting.” arXiv:2505.22837 (2025). [qBraid x Airbus industrial QRC case study.]
6. Hou, Y. et al. “High-Accuracy Temporal Prediction via Experimental Quantum Reservoir Computing in Correlated Spins.” Phys. Rev. Lett. (2026). arXiv:2508.12383. [9-qubit experimental QRC matching a 10,000-node classical reservoir on multi-step weather prediction.]
7. Čindrak, S., Giebeler, L., Götting, N., Gies, C., Lüdge, K. “Memory-Nonlinearity Trade-off across Quantum Reservoir Computing Frameworks.” arXiv:2603.21371 (2026). [Unifies QRC design choices through a memory–nonlinearity tradeoff.]
8. Antoncich, L. et al. “Quantum Reservoir Computing with Neutral Atoms on a Small, Complex, Medical Dataset.” arXiv:2602.14641 (2026). [QRC on QuEra Aquila; observes hardware-induced regularization vs. noiseless emulation.]

4. GUIDELINES AND EXPECTATIONS

Detailed Phase 2 submission requirements, evaluation rubric, and rules are provided below. Teams must adhere to the standard Global Industry Challenge standardized Phase 2 format. Phase 3 will require additional reproducibility deliverables, including an executable qBraid workflow and an agentic reproducibility package; details will be released to finalist teams.

Submission Requirements

Submissions must follow the standardized Phase 2 format: a maximum of **3 pages (PDF)**, excluding the required cover page template and references, using 11-point Times New Roman, single spacing, and submitted via Aqora. Only one submission is required per team.

The **official Global Industry Challenge cover page template is required** and must be included as the first page of the submission. Teams may not recreate, modify, or substitute this template. Submissions that do not comply with these requirements may be disqualified.

Your submission should address:

- Focus area and rationale
- Technical approach to quantum integration
- Stakeholder relevance
- Data modeling strategy
- Quantum platform justification and resource needs

Rules, Eligibility, and Phase 3 Outlook

- All Phase 2 work must use publicly available datasets. No proprietary or restricted data is required at this phase.
- Submissions must be original to the team. Use of published literature, open-source codebases, and large language model assistants is permitted but must be disclosed.
- Phase 3 finalists will execute their proposed architecture, run scaling and noise studies, and benchmark against classical baselines. Final deliverables will include a 5-page technical paper, an executable code repository on qBraid, and an agent-executable reproducibility package.
- Solutions must be reproducible. Code, data references, and run instructions must be sufficient for a third-party reviewer to verify the headline results in Phase 3.
- Teams must disclose any use of proprietary data and cannot submit results that depend on data the judges cannot access.
- Use of generative AI tools is permitted for code support and writing, but the technical contributions, formulations, and results must be the team's own work.

Evaluation Criteria
Submissions will be evaluated by a panel of qBraid, MITRE, JonesTrading, and academic reviewers using the rubric below;

Criterion	Description
1. QRC Architecture Design	Soundness and rigor of the proposed Hamiltonian, encoding, readout, and hybrid integration.
2. Theoretical & Analytical Justification	Quality of the case for QRC on the chosen problem; supporting literature and prototyping evidence.
3. Data Modeling Strategy	Appropriateness of dataset, preprocessing, baselines, and evaluation metrics.
4. Track Selection & Problem Framing	Clarity of the targeted sub-problem and rationale for matching it to a QRC approach.
5. Platform Justification & Resources	Appropriateness of selected backends, qubit/depth/shot estimates, and integration plan.
6. Phase 3 Execution Plan	Concreteness and feasibility of the Phase 3 milestone plan, including fallback options.
7. Clarity of Communication	Structure, coherence, and readability for both quantum and domain experts.

5. SUPPORT AND COMMUNICATION

Mentorship & Office Hours
qBraid will host office hours once every two weeks during Phase 2, covering platform usage, QRC architecture design, and dataset access. A dedicated Discord channel will be made available to all down-selected teams.

Q&A Sessions
A live Q&A webinar will be held within one week of Phase 2 kickoff (target: early May 2026), with a recording posted to the Aqora Challenge page for participants in different time zones.

Contact
For Challenge-specific technical questions, contact the qBraid Challenge team via the Aqora platform messaging system. For platform issues, use the qBraid in-app support channel.

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Phase 2 Detailed Challenge Description

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Dynamic Systems Forecasting: Design and benchmark quantum reservoir computing systems to forecast complex, chaotic time-series data for use cases in financial and environmental domains.

Phase 2 of the Global Industry Challenge focuses on conceptual design and technical justification. Teams will produce a detailed technical paper that makes the case for their proposed QRC architecture and provides a clear plan for Phase 3 execution. The emphasis is on rigor of design and depth of justification rather than final benchmark results. Teams are expected to perform light-touch prototyping during Phase 2: small-scale experiments (for example, 7–12 qubit reservoirs on a simulator) sufficient to substantiate their core design claims. Detailed performance benchmarking, scaling studies, and noise analysis are deferred to Phase 3.

1. TRACK SELECTION AND PROBLEM FRAMING

Each team must select exactly one of the following two tracks for Phase 2 and Phase 3:

Track A: Financial Volatility Prediction.

Stock markets go through calm periods and turbulent ones, and detecting when the market is about to shift from one regime to another is one of the hardest problems in finance. Using publicly available equity market data, design a QRC system that can identify volatility regime shifts and forecast their transitions. Even marginal accuracy improvements in this domain matter for risk management, portfolio allocation, and derivatives pricing.

Track B: Weather Time-Series Forecasting.

Weather is the textbook example of a chaotic dynamical system: small changes in initial conditions lead to wildly different outcomes. Using real-world weather station data (temperature, pressure, humidity, wind), design a QRC system that forecasts atmospheric variables over short horizons. Better short-term forecasts have direct impact on agriculture, energy grid management, logistics, and disaster preparedness.

Teams must articulate why their chosen track is well-matched to a reservoir-computing approach, what specific sub-problem within the track they are targeting (e.g., realized variance forecasting at a specific horizon, or temperature forecasting at a specific lead time), and why the chosen dataset is appropriate.

2. QRC ARCHITECTURE DESIGN

This is the technical core of the submission. Teams must specify and justify each component of their proposed QRC architecture:

- Reservoir Hamiltonian or unitary structure (e.g., transverse-field Ising, Rydberg analog evolution, gate-based ansatz with random fixed parameters).
- Input encoding scheme (amplitude, angle, phase encoding; encoding density; repeated encoding strategies).
- Readout strategy (which observables, expectation values vs. shot-based estimators, polynomial or kernelized readouts).
- Memory / feedback mechanisms, if any (e.g., recurrent feeding of measurements, mid-circuit measurement, classical memory qubits).
- Hybrid classical–quantum integration: how the reservoir output is consumed by a classical regression or classification head.

A clear diagram of the architecture is strongly encouraged.

3. THEORETICAL AND ANALYTICAL JUSTIFICATION

Teams must articulate why their architecture is expected to perform well on the chosen problem. This section is weighted heavily in the rubric. Strong responses will:

- Identify the specific property of the quantum reservoir being exploited (Hilbert space dimensionality, intrinsic nonlinearity from unitary dynamics, memory from non-equilibrium dynamics, dissipation as a feature, etc.).
- Connect this property to the structure of the target signal (long-memory, multi-scale, regime-switching, non-Gaussian, chaotic with a particular Lyapunov spectrum).
- Cite supporting prior work and identify gaps the team intends to address.
- Present preliminary prototyping results from small-scale (for example, 7–12 qubit) experiments that substantiate the core design claims. These do not need to be final benchmark results; the goal is evidence that the proposed direction is viable.

4. DATA MODELING STRATEGY

Teams must describe:

- The exact public dataset(s) chosen, including source, frequency, and time range.
- Preprocessing pipeline (normalization, windowing, feature engineering, train / validation / test splits). Classical baselines that will be benchmarked against in Phase 3.

Recommended baselines, with brief context:

Track A baselines:

- **GARCH**: classical econometric volatility model based on past squared returns and past variance (Bollerslev 1986).
- **ESN** (Echo State Network): the classical analog of QRC, a fixed random recurrent network with a trained linear readout. The most important baseline since beating ESN is what would justify QRC (Jaeger 2001).
- **LSTM**: gated recurrent neural network designed to capture long-range dependencies (Hochreiter & Schmidhuber 1997).

Track B baselines:

- **Persistence**: the trivial baseline (forecast equals current observation). Surprisingly hard to beat at very short horizons.
- **ARIMA**: classical Box–Jenkins linear time-series model (Wikipedia).
- **ESN**: same as Track A. Particularly strong on chaotic time-series such as Mackey–Glass and Lorenz.
- **NWP-style baselines**: Numerical Weather Prediction outputs from physics-based models. Teams can compare to publicly available NWP forecasts (NOAA GFS, ECMWF IFS) rather than running their own (Wikipedia).

Evaluation metrics. Brief context for each:

Track A metrics:

- **RMSE**: Root Mean Squared Error, the standard L2 regression metric.
- **QLIKE** (Quasi-Likelihood Loss): an asymmetric volatility-specific loss recommended by Patton (2011) because RMSE-on-variance can be misleading.
- **Mincer–Zarnowitz regression**: regress realized values on forecasts; test the joint hypothesis (intercept = 0, slope = 1) for forecast unbiasedness and efficiency (Mincer & Zarnowitz 1969).

Track B metrics:

- **RMSE and MAE**: standard regression metrics. MAE is more interpretable in physical units (for example, “off by 2°C on average”).
- **Valid Prediction Time (VPT)**: for chaotic systems, the horizon at which forecast error first exceeds a threshold, typically normalized by the system’s Lyapunov time. Standard in the chaotic-forecasting literature.

5. QUANTUM PLATFORM AND RESOURCE PLANNING

Phase 2 submissions should specify which qBraid-accessible platforms the team intends to use in Phase 3. The qBraid platform provides access to a wide range of simulators (state-vector, density-matrix, tensor-network, and GPU-accelerated) and quantum hardware backends from major providers (Amazon Braket, IBM, IonQ, IQM, QuEra, Rigetti, and others). Teams may also access error mitigation tooling such as Mitiq, mthree, and Zero-Noise Extrapolation (ZNE) when targeting noisy hardware.

Teams must justify their platform selection(s) and provide concrete technical estimates: expected qubit count range, circuit depth, simulation parameters, shot budgets, and integration with classical infrastructure. Simulator-first plans are encouraged, with QPU access used for targeted scaling and validation experiments.

6. STAKEHOLDER IMPACT

Briefly describe who would benefit from a successful solution. For Track A, this might include trading desks, risk managers, and market makers; for Track B, emergency-response agencies, energy grid operators, agriculture, and aviation. Outline a milestone plan for Phase 3 execution: what experiments will be run, in what order, and what fallback options exist if scaling hits hardware or simulator limits.